

An Analysis of Recommender system

Recommender system

Recommender system -- Part 1

Performance measures Evaluation is important in assessing the effectiveness of recommendation algorithms. To measure the effectiveness of recommender systems, and compare different approaches, three types of evaluations are available: user studies, online evaluations (A/B tests), and offline evaluations. The commonly used metrics are the mean squared error and root mean squared error, the latter having been used in the Netflix Prize. The information retrieval metrics such as precision and recall or discounted cumulative gain (DCG) are useful to assess the quality of a recommendation method. Diversity, novelty, and coverage are also considered as important aspects in evaluation. However, many of the classic evaluation measures are highly criticized. Evaluating the performance of a recommendation algorithm on a fixed test dataset will always be extremely challenging as it is impossible to accurately predict the reactions of real users to the recommendations. Hence any metric that computes the effectiveness of an algorithm in offline data will be imprecise. User studies are rather a small scale. A few dozens or hundreds of users are presented recommendations created by different recommendation approaches, and then the users judge which recommendations are best. In A/B tests, recommendations are shown to typically thousands of users of a real product, and the recommender system randomly picks at least two different recommendation approaches to generate recommendations. The effectiveness is measured with implicit measures of effectiveness such as conversion rate or click-through rate. Offline evaluations are based on historic data, e.g. a dataset that contains information about how users previously rated movies. The effectiveness of recommendation approaches is then measured based on how well a recommendation approach can predict the users' ratings in the dataset. While a rating is an explicit expression of whether a user liked a movie, such information is not available in all domains. For instance, in the domain of citation recommender systems, users typically do not rate a citation or recommended article. In such cases, offline evaluations may use implicit measures of effectiveness. For instance, it may be assumed that a recommender system is effective that is able to recommend as many articles as possible that are contained in a research article's reference list. However, this kind of offline evaluations is seen critical by many researchers. For instance, it has been shown that results of offline evaluations have low correlation with results from user studies or A/B tests. A dataset popular for offline evaluation has been shown to contain duplicate data and thus to lead to wrong conclusions in the evaluation of algorithms. Often, results of so-called offline evaluations do not correlate with actually assessed user-satisfaction. This is probably because offline training is highly biased toward the highly reachable items, and offline testing data is highly influenced by the outputs

of the online recommendation module. Researchers have concluded that the results of offline evaluations should be viewed critically. Researchers have identified limits to the practical usefulness of accuracy-based evaluation metrics in recommender systems, sometimes described as an accuracy barrier or magic barrier. In this view, continued improvements in offline accuracy do not necessarily translate into better user experience, satisfaction, or real-world system impact. As a result, accuracy gains beyond a certain point may provide diminishing returns when evaluated solely through offline experiments.

Recommender system -- Part 2

Mobile recommender systems make use of internet-accessing smartphones to offer personalized, context-sensitive recommendations. This is a particularly difficult area of research as mobile data is more complex than data that recommender systems often have to deal with. It is heterogeneous, noisy, requires spatial and temporal auto-correlation, and has validation and generality problems. There are three factors that could affect the mobile recommender systems and the accuracy of prediction results: the context, the recommendation method and privacy. Additionally, mobile recommender systems suffer from a transplantation problem – recommendations may not apply in all regions (for instance, it would be unwise to recommend a recipe in an area where all of the ingredients may not be available). One example of a mobile recommender system are the approaches taken by companies such as Uber and Lyft to generate driving routes for taxi drivers in a city. This system uses GPS data of the routes that taxi drivers take while working, which includes location (latitude and longitude), time stamps, and operational status (with or without passengers). It uses this data to recommend a list of pickup points along a route, with the goal of optimizing occupancy times and profits.

Recommender system -- Part 3

Risk-aware recommender systems The majority of existing approaches to recommender systems focus on recommending the most relevant content to users using contextual information, yet do not take into account the risk of disturbing the user with unwanted notifications. It is important to consider the risk of upsetting the user by pushing recommendations in certain circumstances, for instance, during a professional meeting, early morning, or late at night. Therefore, the performance of the recommender system depends in part on the degree to which it has incorporated the risk into the recommendation process. One option to manage this issue is DRARS, a system which models the context-aware recommendation as a bandit problem. This system combines a content-based technique and a contextual

bandit algorithm.

Recommender system -- Part 4

Hybrid recommendations approaches Most recommender systems now use a hybrid approach, combining collaborative filtering, content-based filtering, and other approaches. E-commerce platforms frequently use hybrid approaches to overcome problems like the cold start problem, where a new user has no history for collaborative filtering to analyze. There is no reason why several different techniques of the same type could not be hybridized. Hybrid approaches can be implemented in several ways: by making content-based and collaborative-based predictions separately and then combining them; by adding content-based capabilities to a collaborative-based approach (and vice versa); or by unifying the approaches into one model. Several studies that empirically compared the performance of the hybrid with the pure collaborative and content-based methods and demonstrated that the hybrid methods can provide more accurate recommendations than pure approaches. These methods can also be used to overcome some of the common problems in recommender systems such as cold start and the sparsity problem, as well as the knowledge engineering bottleneck in knowledge-based approaches. Netflix is a good example of the use of hybrid recommender systems. The website makes recommendations by comparing the watching and searching habits of similar users (i.e., collaborative filtering) as well as by offering movies that share characteristics with films that a user has rated highly (content-based filtering). Some hybridization techniques include: